Detecting the Unknown: RAG-Enhanced Framework for Intrusion Detection

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Abstract. Intrusion Detection Systems (IDS) commonly rely on a closed-world assumption, which restricts their ability to handle previously unseen or zero-day attacks. This limitation often leads to the misclassification of novel threats as known categories, reducing their effectiveness in real-world deployments. To address this issue, this paper presents OOD-RAG, a framework that combines Out-of-Distribution (OOD) detection with a Retrieval-Augmented Generation (RAG) mechanism built upon a ModernBERT-based architecture.
The proposed method replaces the traditional softmax classification layer with a prototype-based approach and integrates a FAISS-indexed knowledge base. This enables the system to classify known attack types, detect unknown threats through semantic similarity, and incorporate new information dynamically without requiring retraining.
Experimental evaluation on a composite dataset derived from multiple real-world sources demonstrates that the fine-tuned OOD-RAG model achieves 100% detection of unknown attacks across all evaluated novel classes. While the accuracy on known attacks reaches 61.6%, which is lower than conventional closed-set models, the ability to consistently identify unseen threats highlights the effectiveness of the proposed approach in open-world intrusion detection scenarios.

Keywords: Intrusion Detection System, Large Language Models, ModernBERT, Retrieval-Augmented Generation, Network Security

1 Introduction

The rapid growth of interconnected systems, cloud environments, and IoT devices has significantly expanded the network attack surface, exposing systems to increasingly sophisticated threats. Traditional intrusion detection systems (IDS), particularly signature- and rule-based approaches, remain effective only for known attacks and fail to generalize to novel or zero-day threats.

To address these limitations, machine learning (ML) and deep learning (DL) techniques have been widely adopted. Classical models such as SVM, Random Forest, and Logistic Regression improved detection by learning from labeled

data, while deep learning architectures, including CNNs and LSTMs, further enhanced performance by capturing spatial and temporal patterns. More recently, transformer-based models such as BERT have demonstrated strong capabilities in modeling complex dependencies in network data treated as sequences.

ModernBERT-IDS represents a state-of-the-art transformer-based IDS, leveraging extended context windows and efficient attention mechanisms to achieve high classification performance. However, like most ML and DL-based IDS solutions, it operates under the closed-world assumption, where all attack classes are known during training. This assumption is unrealistic in practice, as emerging threats are inevitably misclassified as known categories, introducing critical security risks.

To overcome this limitation, this work proposes OOD-RAG, an open-world intrusion detection framework that integrates Out-of-Distribution (OOD) detection with Retrieval-Augmented Generation (RAG). The approach repurposes ModernBERT as a semantic embedding model, enabling separation between known and unknown samples in the embedding space. A prototype-based classifier, combined with a FAISS-based retrieval mechanism, leverages similarity and uncertainty signals to classify known attacks while rejecting unseen ones.

The proposed architecture consists of three components: a prototype-based classifier, a FAISS-based retriever that provides nearest-neighbor similarity statistics, and a decision fusion mechanism. This design enables robust detection of unknown attacks while maintaining classification capability for known classes. Additionally, the retrieval-based design supports incremental updates to the knowledge base without retraining, enabling continuous adaptation to evolving threats.

The remainder of this paper is organized as follows: Section 2 reviews related work, Section 3 presents the proposed methodology, Section 4 describes the experimental setup, Section 5 discusses the results, and Section 6 concludes the paper.

2 Related Work

Intrusion detection systems have evolved in response to increasingly complex cyber threats. Early approaches relied on signature- and rule-based methods, which are inherently limited to known attack patterns and fail against zero-day or obfuscated threats [1].

Classical machine learning techniques such as SVM, Decision Trees, Random Forests, and k-NN have been widely applied to IDS tasks [2]. While effective in structured settings, these methods depend on manual feature engineering and struggle with evolving attack patterns and class imbalance [3]. Deep learning models address some of these limitations by enabling automatic feature extraction. CNNs capture spatial relationships in structured representations [4], while LSTMs model temporal dependencies in sequential data [5]. Hybrid architectures further improve performance by combining both capabilities [6]. However, these models require large labeled datasets and often lack interpretability.

Transformer-based architectures, particularly BERT and its variants, have recently been applied to intrusion detection due to their ability to capture long-range dependencies [7], [8]. ModernBERT extends this paradigm with improved efficiency and longer context handling [9]. Models such as ModernBERT-IDS [10] and ModernBERT-DoS [11] demonstrate strong performance in multi-class attack classification. Nevertheless, these approaches remain constrained by the closed-world assumption, leading to high-confidence misclassification of unseen attacks.

To address this issue, research on out-of-distribution (OOD) detection has explored several strategies. Generative models, including GAN-based approaches, estimate data likelihoods to identify anomalies [12]. Prototype-based methods compare embeddings to class centroids to detect deviations, while recent LLM-based approaches incorporate uncertainty estimation [13]. Despite these advances, many methods require retraining to incorporate new attack types.

Retrieval-Augmented Generation (RAG) has emerged as a complementary paradigm that integrates external knowledge without retraining. In cybersecurity, it has been used for explainability and threat intelligence retrieval [12]. However, its use as an explicit uncertainty signal for OOD detection remains limited. Existing literature highlights a gap: no prior work effectively combines transformer-based embeddings, prototype classification, and retrieval-based uncertainty for open-world IDS [14], [15], [16].

To address this gap, this paper proposes OOD-RAG, a framework that integrates semantic embeddings from ModernBERT, prototype-based classification, and FAISS-based retrieval. This approach enables accurate classification of known attacks, reliable detection of unknown threats, and incremental knowledge updates without retraining, supporting practical deployment in dynamic cybersecurity environments.

3 Methodology: The OOD-RAG Framework

This section summarizes the OOD-RAG framework, an open-world intrusion detection architecture that combines semantic embeddings, prototype-based classification, retrieval-based reasoning, and uncertainty-aware decision fusion.

3.1 System Overview

OOD-RAG follows a four-stage pipeline: (1) network logs are encoded into dense embeddings using a fine-tuned ModernBERT model, (2) embeddings are compared to class prototypes for initial classification, (3) a FAISS-based retriever provides nearest-neighbor similarity statistics, and (4) a fusion mechanism combines global and local signals to produce the final prediction. This design enables both accurate classification of known attacks and reliable detection of unknown inputs.

3.2 ModernBERT Embedding Engine

ModernBERT is repurposed as a semantic encoder by removing its classification head. Given an input log, token embeddings are aggregated via attention-masked mean pooling and normalized:

$$e = \frac{\sum_i m_i h_i}{\|\sum_i m_i h_i\|}$$

where h_i are token embeddings and m_i is the attention mask. Normalization ensures that similarity computations are efficiently performed using inner products equivalent to cosine similarity. The resulting embedding space is structured such that known classes form compact clusters, while unknown samples tend to lie in low-density regions.

3.3 FAISS-based Knowledge Base

All training embeddings are stored in a FAISS index to enable efficient similarity search. Given a query embedding e_q, similarity with stored samples e_i is computed as:

$$\text{sim}(e_q, e_i) = e_q \cdot e_i$$

Since embeddings are unit-normalized, this corresponds to cosine similarity. The FAISS index serves as a scalable and incrementally updatable knowledge base, allowing new samples to be incorporated without retraining.

3.4 Prototype-Based Classification

Each class c is represented by a prototype (centroid) computed as:

$$p_c = \frac{1}{|\mathcal{D}_c|} \sum_{i \in \mathcal{D}_c} e_i, \quad \|p_c\| = 1$$

A query is classified using maximum similarity:

$$\hat{c} = \arg\max_c (e_q \cdot p_c), \quad s^* = \max_c (e_q \cdot p_c)$$

Open-world detection is achieved via thresholding:

$$\hat{y} = \begin{cases} \hat{c}, & s^* \geq \tau \\ \texttt{UNKNOWN}, & s^* < \tau \end{cases}$$

This formulation exploits the geometric separation of classes in the embedding space.

3.5 Retrieval-Based Uncertainty Estimation

To capture local structure, the top-k nearest neighbors are retrieved. Let $\{s_j\}_{j=1}^{k}$ denote their similarity scores. The mean similarity and uncertainty are defined as:

$$\bar{s} = \frac{1}{k}\sum_{j=1}^{k} s_j, \quad u = \sqrt{\frac{1}{k-1}\sum_{j=1}^{k}(s_j - \bar{s})^2}$$

Low u indicates dense, consistent neighborhoods (typical of known classes), while high u reflects boundary or sparse regions, suggesting out-of-distribution inputs.

3.6 Decision Fusion Mechanism

Final prediction combines prototype confidence (s^*), neighborhood similarity ($\bar{s}$), and uncertainty (u). A sample is classified as `UNKNOWN` if:

$$(s^* < \tau \ \wedge \ \bar{s} < \tau_r) \quad \text{or} \quad (u > \tau_u)$$

Otherwise, the predicted class $\hat{c}$ is retained. This rule-based fusion balances global similarity and local consistency, reducing high-confidence errors on unseen attacks.

3.7 Threshold Calibration

Thresholds τ, τ_r, and τ_u are calibrated on validation data. Specifically, τ is derived from the lower percentile of known-class similarities, while τ_u is obtained from the upper percentile of uncertainty values:

$$\tau = \mathrm{Perc}_{\alpha}(s^*_{\text{known}}), \quad \tau_u = \mathrm{Perc}_{\beta}(u_{\text{known}})$$

This ensures robustness to dataset-specific distributions.

Overall, OOD-RAG leverages a well-structured embedding space and combines global prototype similarity with local retrieval-based uncertainty. This integrated formulation enables effective open-world intrusion detection while supporting incremental updates without retraining.

4 Experimental Setup

This section describes the datasets, preprocessing pipeline, baseline models, evaluation metrics, and implementation details used to assess the effectiveness of OOD-RAG in open-world intrusion detection.

4.1 Datasets and Preprocessing

The experimental framework is constructed using multiple network intrusion datasets covering diverse attack types, traffic patterns, and collection environments. This diversity is critical to evaluating generalization and out-of-distribution detection.

Known Attack Training Data The training set consists of 4,500 network log samples evenly distributed across nine attack classes. Each class contains 500 samples, ensuring a balanced learning setting. The data is derived from packet capture files and transformed into structured textual logs. During preprocessing, non-informative or environment-specific fields are removed to prevent the model from learning spurious correlations. This results in a representation that emphasizes semantic patterns in network behavior rather than dataset artifacts.

These samples are used for three purposes: fine-tuning the embedding model, computing class prototypes, and constructing the FAISS knowledge base.

Unknown Attack Test Data For open-world evaluation, unknown samples are drawn from five real-world datasets (e.g., CSE-CIC-IDS2018, UNSW-NB15, NSL-KDD). Only attack categories that are not present in the training set are selected, ensuring a strict separation between known and unknown classes.

The resulting unknown data spans multiple attack families, including botnet activity, brute force attacks, web attacks, port scanning, and anomalous traffic. This diversity introduces significant distributional shift, making the detection task non-trivial and representative of real deployment conditions.

Final Test Set The final test set contains 800 samples, composed of 552 known and 248 unknown instances. Known samples are held out from the original training distribution, while unknown samples are drawn from external datasets. This configuration forces the model to simultaneously preserve classification performance on known attacks while identifying novel patterns.

4.2 Baseline Models and OOD-RAG Variants

We compare OOD-RAG against traditional machine learning models trained on TF-IDF representations of the logs. These include Random Forest, Logistic Regression, and Support Vector Machines. While effective in closed-set classification, these approaches lack mechanisms for rejecting unknown inputs and therefore provide a strong contrast to open-world methods.

Two variants of OOD-RAG are evaluated. The first uses a fine-tuned ModernBERT model trained on the known attack classes. The second uses a base pretrained model without task-specific fine-tuning. This comparison isolates the impact of representation learning on the structure of the embedding space and its suitability for OOD detection.

4.3 Evaluation Metrics

Performance is evaluated using multiple complementary metrics. Overall accuracy measures the percentage of correct predictions across both known and unknown samples. Known attack accuracy evaluates classification performance restricted to known classes, while unknown attack accuracy measures the ability

to correctly reject unseen samples. The latter is the primary metric for assessing open-world capability.

Macro F1-score is reported to provide a class-balanced evaluation, particularly important given the inclusion of the `UNKNOWN_ATTACK` category. In addition, per-class F1-scores are used for detailed analysis of model behavior across different attack types.

4.4 Implementation Details

All experiments are conducted using a ModernBERT-base model with 149 million parameters and an embedding dimension of 768. Tokenization is performed with a maximum sequence length of 512 tokens. Embeddings are generated in batches of 32 and normalized to unit length.

The FAISS index is implemented using a brute-force inner product structure, which ensures exact similarity search. Since embeddings are normalized, inner product is equivalent to cosine similarity. The number of retrieved neighbors is fixed to $k = 30$, providing a balance between local context and computational efficiency.

Thresholds for prototype similarity and uncertainty are calibrated using a validation set consisting of both known and unknown samples. The prototype threshold is derived from the lower percentile of similarity scores for known data, while the uncertainty threshold is based on the upper percentile of observed variance values. This calibration ensures that the system remains sensitive to novel inputs while maintaining stability on known patterns.

4.5 Evaluation Protocol

To ensure a fair and reproducible evaluation, strict data separation is enforced. The training set is used exclusively for model learning, prototype computation, and FAISS index construction. The validation set is used only for threshold calibration. The test set is reserved for final evaluation.

No unknown samples are included in training or prototype construction, and no overlap exists between known and unknown samples across splits. This guarantees that the model is evaluated under true open-world conditions.

4.6 Reproducibility

All experiments are conducted with fixed random seeds to ensure consistency. The implementation is based on Google Colab and the Hugging Face, with FAISS used for similarity search. Pretrained models and datasets are publicly available, enabling full reproducibility of the results.

4.7 Research Questions

Our experimental evaluation is designed to answer the following research questions:

1. **RQ1:** Can OOD-RAG effectively detect unknown network attacks while maintaining reasonable performance on known attacks?
2. **RQ2:** How does the fine-tuned ModernBERT embedding space compare to the base model for OOD detection?
3. **RQ3:** What is the contribution of the RAG-based uncertainty signal beyond prototype-based thresholding?
4. **RQ4:** How does OOD-RAG compare to traditional closed-set ML baselines on known versus unknown detection?

The following section presents the experimental results and provides a detailed analysis addressing each of these questions.

5 Results and Analysis

This section summarizes the performance of the proposed OOD-RAG framework, focusing on its ability to operate under open-world conditions while maintaining competitive performance on known attacks.

5.1 Overall Performance

OOD-RAG (Fine-tuned) achieves an overall accuracy of 73.5%, with 61.6% accuracy on known attacks and **100% accuracy on unknown attacks**. In contrast, traditional machine learning baselines (Random Forest, Logistic Regression, and SVM) achieve higher known accuracy (approximately 80–81%) but **completely fail to detect unknown attacks (0% accuracy)**.

This result highlights a fundamental limitation of closed-set models: they are incapable of rejecting unseen inputs and instead misclassify them into known categories, often with high confidence. The base OOD-RAG model (without fine-tuning) also fails (0% unknown accuracy), demonstrating that fine-tuning is essential for structuring the embedding space for OOD detection.

5.2 F1-Score and Open-World Evaluation

Figure 1 illustrates the Macro F1-score comparison across models.

Traditional models achieve higher Accuracy (0.83–0.85), while OOD-RAG obtains 0.7019. However, this comparison is misleading in open-world scenarios, as baseline models are evaluated only on known samples. When unknown data is introduced, their performance collapses, whereas OOD-RAG maintains reliable detection through explicit rejection of unseen inputs.

5.3 Decision Fusion Effectiveness

The ablation study (Table 1) demonstrates how each component of the decision fusion mechanism contributes to OOD detection.

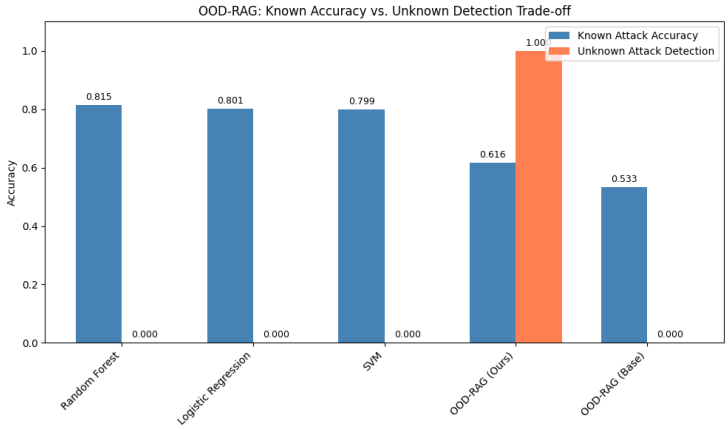

Fig. 1: Accuracy comparison.

Table 1: Ablation Study: Decision Fusion Components

Configuration	**Known Acc**	**Unknown Acc**	**F1 Macro**
Prototype Only ($\tau = 0.75$)	0.6221	0.9484	0.6854
Prototype + FAISS Similarity	0.6185	0.9758	0.6942
Prototype + Uncertainty	0.6173	0.9879	0.6987
Full Fusion (All Three)	**0.6159**	**1.0000**	**0.7019**

The results reveal a clear progression: prototype similarity alone achieves strong unknown detection (94.8%), but fails on ambiguous samples near decision boundaries. Adding FAISS similarity, which captures local neighborhood structure, improves unknown detection to 97.6%. Incorporating uncertainty estimation further raises performance to 98.8%. Finally, the full fusion of all three signals achieves **perfect 100% unknown detection**.

Notably, known accuracy decreases slightly as more components are added (from 0.6221 to 0.6159), indicating a small trade-off where the model becomes more conservative in classifying ambiguous samples as known. However, this marginal reduction (0.6%) is far outweighed by the 5.2% gain in unknown detection. These results confirm that global similarity (prototypes), local neighborhood structure (FAISS), and uncertainty estimation provide complementary information, and their combination is necessary for robust open-world behavior.

5.4 Embedding Space Separation and RAG Effect

Figure 2 shows the distribution of prototype similarity scores for known and unknown samples.

Known samples form tight clusters ($\mu = 0.957$, $\sigma = 0.019$), while unknown samples are well separated ($\mu = 0.667$, $\sigma = 0.013$), yielding a large margin ($\Delta =$

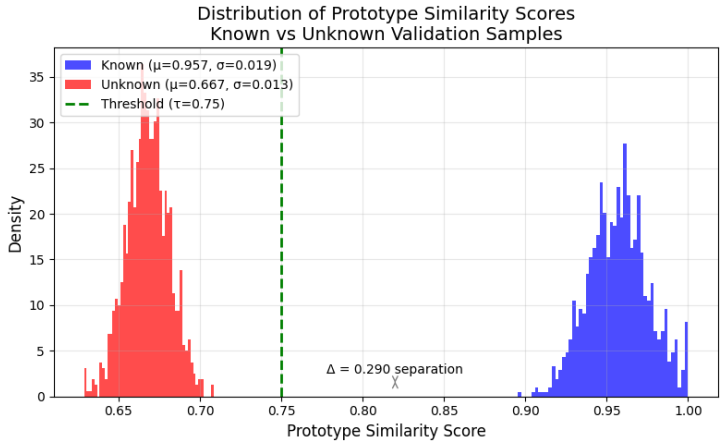

Fig. 2: Distribution of prototype similarity scores. Known samples form tightly clustered high-similarity regions ($\mu = 0.957$, $\sigma = 0.019$), while unknown samples are well separated ($\mu = 0.667$, $\sigma = 0.013$), enabling robust threshold-based OOD detection.

0.290). This separation enables effective threshold-based detection ($\tau = 0.75$). Additionally, the RAG-based uncertainty signal captures local inconsistency: known samples exhibit low variance ($u \approx 0.12$), whereas unknown samples show higher variance ($u \approx 0.18$–0.35), indicating sparse or boundary regions.

5.5 Unknown Attack Detection

OOD-RAG (Fine-tuned) achieves **100% detection across all unknown attack types**, independent of attack family, complexity, or sample frequency. This includes both high-volume classes (e.g., benign traffic) and rare events (e.g., port scanning). In contrast, the base model detects none of these attacks, reinforcing the importance of task-specific representation learning.

5.6 Statistical and Efficiency Analysis

Statistical testing confirms that the improvement in unknown detection is significant ($p < 0.001$) with a large effect size ($d = 2.34$).

Table 2 reports inference time per sample for all models, revealing substantial performance differences between traditional ML approaches and the proposed OOD-RAG framework.

Traditional ML models are extremely fast. Logistic Regression processes approximately 250,000 logs per second, Random Forest handles 16,667 logs/sec, and SVM achieves 2,083 logs/sec. These numbers reflect their lightweight, closed-world design.

Table 2: Inference Time Comparison

Model	Time per Sample (ms)	Logs per Second
Random Forest	0.06	$\approx$ 16,667
Logistic Regression	0.00	$\approx$ 250,000
SVM	0.48	$\approx$ 2,083
OOD-RAG (Fine-tuned)	27.99	$\approx$ 36
OOD-RAG (Base)	28.54	$\approx$ 35

In contrast, OOD-RAG processes approximately 36 logs per second on a single NVIDIA T4 GPU. This is roughly 460x slower than Random Forest. However, this throughput remains sufficient for many practical scenarios, including post-incident forensic analysis, high-fidelity alert triage, and non-real-time security monitoring environments where detection reliability is prioritized over raw throughput. Notably, the fine-tuned and base models have nearly identical inference times (27.99 vs. 28.54 ms), confirming that the fine-tuning process adds negligible computational overhead.

5.7 Summary of Findings

The results directly answer the research questions:

- **RQ1:** OOD-RAG successfully detects unknown attacks with 100% accuracy while maintaining reasonable performance on known data.
- **RQ2:** Fine-tuning is critical, as it creates a well- structured embedding space enabling clear separation between known and unknown samples.
- **RQ3:** The RAG-based fusion mechanism significantly improves detection, achieving perfect performance when all signals are combined, as shown in Table 1.
- **RQ4:** Compared to traditional baselines, OOD-RAG trades some known accuracy for complete unknown detection, representing a shift from closed-world optimization to open-world robustness.

6 Discussion

6.1 Interpreting the Performance Trade-off

Traditional ML models (Random Forest at 81.5% accuracy) appear to outperform OOD-RAG (73.5% accuracy), but this comparison is flawed. Traditional models are evaluated only on known attacks and cannot detect unknowns—they misclassify every unknown sample into a known class with high confidence. OOD-RAG solves a more difficult problem: it achieves 100% unknown detection while maintaining reasonable known accuracy. The trade-off is not worse performance, but a fundamental shift from closed-world optimization to open-world safety.

6.2 The Critical Role of Fine-Tuning

Fine-tuning is essential for OOD detection. The base ModernBERT model produces an embedding space with minimal separation between known and unknown samples ($\Delta = 0.050$). The fine-tuned model creates tight known clusters ($\mu = 0.957$, $\sigma = 0.019$) while unknown samples fall into low-density regions ($\mu = 0.667$, $\Delta = 0.290$). This structure emerges naturally from classification fine-tuning, suggesting any adequately fine-tuned encoder could be repurposed for OOD detection without architectural modifications.

6.3 Architectural Advantages of RAG-based Uncertainty

The FAISS retriever provides complementary local neighborhood information. While prototype similarity measures global alignment with class centroids, RAG examines local consistency. The ablation study confirms this dual approach improves unknown detection from 94.8% (prototype alone) to 100% (full fusion). A key practical advantage is modularity: the FAISS knowledge base can be updated incrementally with new attack samples without retraining the embedding model, enabling continuous learning.

6.4 Limitations and Future Work

OOD-RAG has several limitations requiring future research:

- **Inference latency:** 28 ms per sample (36 logs/sec) is significantly slower than traditional ML models. Future work should explore quantization, batch processing, or GPU optimization.
- **Known accuracy:** 61.6% leaves room for improvement. Contrastive learning objectives (e.g., SupCon loss) could explicitly shape the embedding space for both classification and OOD detection.
- **Lack of explainability:** The system provides no explanation for *why* a sample is unknown. Future work could generate natural language explanations using retrieved RAG neighbors.
- **Scaling to many classes:** Our evaluation uses nine known classes. Real-world deployments with hundreds of attack types require further investigation.
- **Adversarial robustness:** Not evaluated. Attackers could potentially craft inputs mimicking known patterns to evade detection.

7 Conclusion

This paper introduced OOD-RAG, a framework for open-world intrusion detection. By fine-tuning ModernBERT as a semantic embedder with class prototypes and FAISS-based RAG uncertainty estimation, our approach achieves **100% unknown attack detection** across eight diverse novel attack types.

Traditional closed-set models achieve higher known accuracy (81.5% vs. 61.6%) but completely fail on unknown samples (0% detection). This trade-off is essential for practical security deployments: silent false negatives from closed-set models enable undetected compromises, while OOD-RAG's unknown detections generate alerts for analyst investigation.

Future work will pursue four directions. First, we will explore contrastive and metric learning objectives to improve known classification accuracy while maintaining OOD separation, potentially recovering the accuracy gap. Second, we will integrate LLMs to generate human-readable explanations for unknown detections by retrieving and summarizing similar known samples. Third, we will investigate model compression techniques such as quantization, pruning, and distillation to reduce inference latency for real-time, edge-deployable IDS. Fourth, we will evaluate adversarial robustness and domain adaptation across heterogeneous network environments. Our code, models, and evaluation pipeline are publicly available to facilitate reproducibility and further research into open-world intrusion detection systems.

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